

## Q1

(i)

$$y = \frac{1}{4x - x^2} = (4x - x^2)^{-1}$$

$$\frac{dy}{dx} = -(4x - x^2)^{-2} (4 - 2x) = \frac{2x - 4}{(4x - x^2)^2}$$

At turning point,

$$\frac{dy}{dx} = -(4x - x^2)^{-2} (4 - 2x) = 0 \Rightarrow x = 2$$

$$\frac{d^2y}{dx^2} = -\left[(4x - x^2)^{-2}(-2) - 2(4x - x^2)^{-3}(4 - 2x)^2\right]$$

$$\text{When } x = 2, \frac{d^2y}{dx^2} = -\left[(4(2) - (2)^2)^{-2}(-2)\right] = \frac{1}{8} > 0$$

Or

Using first derivative test,

|                 |         |   |        |
|-----------------|---------|---|--------|
| $x$             | 1.9     | 2 | 2.1    |
| $\frac{dy}{dx}$ | -0.0126 | 0 | 0.0126 |
| tangent         | \       | - | /      |

The turning point at  $x = 2$  is a minimum point.

(ii)

Area

$$= \int_1^2 \frac{1}{4x - x^2} dx$$

$$= \int_1^2 \frac{1}{-(x^2 - 4x + 2^2 - 2^2)} dx$$

$$= \int_1^2 \frac{1}{2^2 - (x - 2)^2} dx$$

$$= \left[ \frac{1}{2(2)} \ln \left| \frac{2 + (x - 2)}{2 - (x - 2)} \right| \right]_1^2$$

$$= \left[ \frac{1}{4} \ln \left| \frac{x}{4 - x} \right| \right]_1^2$$

$$= -\frac{1}{4} \ln \frac{1}{3}$$

$$= \frac{1}{4} \ln 3$$

or

$$= \int_1^2 \frac{1}{x(4 - x)} dx$$

$$= \frac{1}{4} \int_1^2 \frac{1}{x} + \frac{1}{4 - x} dx$$

$$= \frac{1}{4} [\ln|x| - \ln|4 - x|]_1^2$$

$$= \frac{1}{4} \left[ \ln \left| \frac{x}{4 - x} \right| \right]_1^2$$

$$= -\frac{1}{4} \ln \frac{1}{3}$$

$$= \frac{1}{4} \ln 3$$

Q2

(i)

$$\frac{1}{(x-a)^2} = |x-a|$$

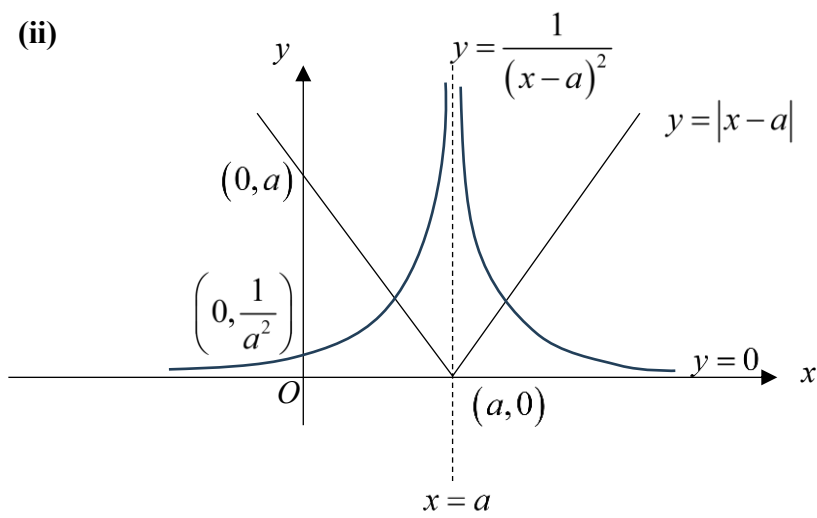
$$\frac{1}{(x-a)^2} = x-a \quad \text{or} \quad \frac{1}{(x-a)^2} = -(x-a)$$

$$(x-a)^3 = 1 \quad (x-a)^3 = -1$$

$$x-a = 1 \quad x-a = -1$$

$$x = a+1 \quad x = a-1$$

(ii)



For  $\frac{1}{(x-a)^2} > |x-a|$ ,

$$a-1 < x < a+1, \quad x \neq a$$

Or

$$a-1 < x < a \quad \text{or} \quad a < x < a+1$$

**Q3**

Let  $h$  m be the vertical distance between the top of the ladder and the floor

Let  $x$  m be the horizontal distance between the foot of the ladder and the corner of the wall

**Method 1**

By Pythagoras' theorem,

$$h^2 + x^2 = 3.12^2$$

By implicit differentiation w.r.t.  $t$ ,

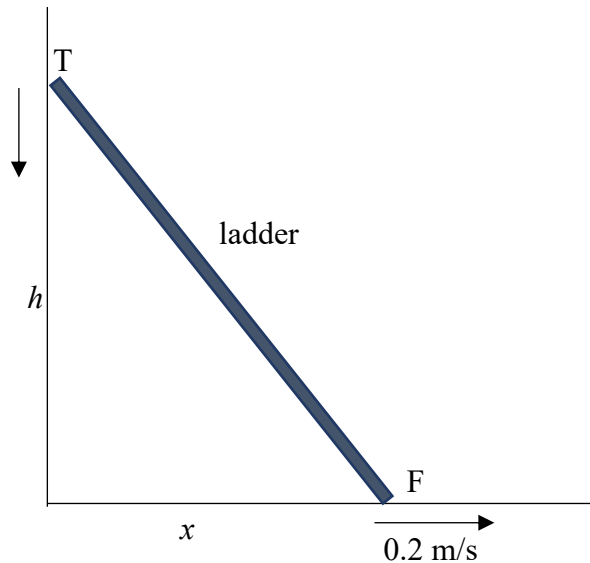
$$2h \frac{dh}{dt} + 2x \frac{dx}{dt} = 0$$

When  $h = 1.2$ ,  $\frac{dx}{dt} = 0.2$  and

$$\begin{aligned} x^2 &= 3.12^2 - 1.2^2 \\ x &= 2.88 \quad (\text{Since } x > 0) \end{aligned}$$

Hence,

$$\begin{aligned} 2(1.2) \frac{dh}{dt} + 2(2.88)(0.2) &= 0 \\ \frac{dh}{dt} &= -0.48 \end{aligned}$$



Hence, the top of ladder is sliding down at a rate of 0.48 m/s.

**Method 2**

By Pythagoras' theorem,

$$h^2 + x^2 = 3.12^2$$

By implicit differentiation w.r.t.  $x$ ,

or By implicit differentiation w.r.t.  $h$

$$2h \frac{dh}{dx} + 2x = 0 \Rightarrow \frac{dh}{dx} = -\frac{x}{h}$$

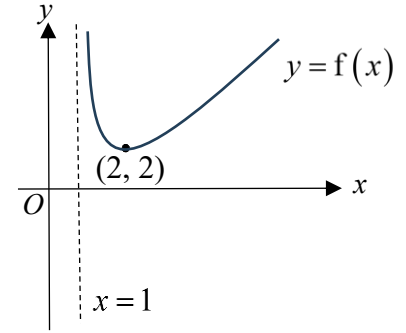
When  $h = 1.2$ ,  $\frac{dx}{dt} = 0.2$  and

$$\begin{aligned} x^2 &= 3.12^2 - 1.2^2 \\ x &= 2.88 \quad (\text{Since } x > 0) \end{aligned}$$

Hence,  $\frac{dh}{dt} = \frac{dh}{dx} \times \frac{dx}{dt}$

$$\begin{aligned} &= -\frac{2.88}{1.2} \times 0.2 \\ &= -0.48 \end{aligned}$$

Hence, the top of ladder is sliding down at a rate of 0.48 m/s.

**Q4****(i)**Smallest value of  $a = 2$ .(For  $f^{-1}$  to exist,  $f$  must be a one-one function.)**(ii)**Let  $y = 4 + 3x - x^2$ 

$$y = -\left[x^2 - 3x - 4\right]$$

$$= -\left[\left(x - \frac{3}{2}\right)^2 - \left(-\frac{3}{2}\right)^2 - 4\right]$$

$$= -\left[\left(x - \frac{3}{2}\right)^2 - \frac{25}{4}\right]$$

$$y = -\left(x - \frac{3}{2}\right)^2 + \frac{25}{4}$$

$$\text{or } x^2 - 3x - 4 + y = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-4 + y)}}{2}$$

$$x = \frac{3 \pm \sqrt{25 - 4y}}{2}$$

$$x = \frac{3}{2} \pm \sqrt{\frac{25}{4} - y}$$

$$\left(x - \frac{3}{2}\right)^2 = \frac{25}{4} - y$$

$$x - \frac{3}{2} = \pm \sqrt{\frac{25}{4} - y}$$

$$x = \frac{3}{2} \pm \sqrt{\frac{25}{4} - y}$$

$$\text{since } x \leq \frac{3}{2}, x = \frac{3}{2} - \sqrt{\frac{25}{4} - y}$$

$$\text{Thus, } g^{-1} : x \mapsto \frac{3}{2} - \sqrt{\frac{25}{4} - x}, \quad x \leq \frac{25}{4}$$

$$D_{g^{-1}} = R_g = \left(-\infty, \frac{25}{4}\right]$$

**(iii)**

$$R_{f^{-1}} = D_f = [2, \infty)$$

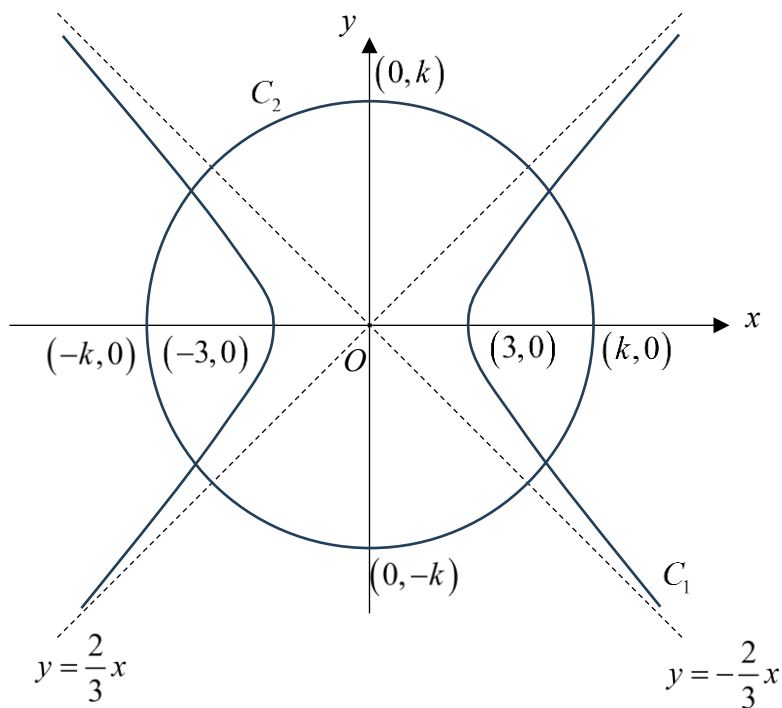
$$D_{g^{-1}} = R_g = \left(-\infty, \frac{25}{4}\right]$$

Since  $R_{f^{-1}} \not\subset D_{g^{-1}}$ ,  $g^{-1}f^{-1}$  does not exist.

Q5

(a)

(i)



(ii)

For  $C_1$  and  $C_2$  to intersect,  $k \geq 3$  (since  $k > 0$ ).

(iii)

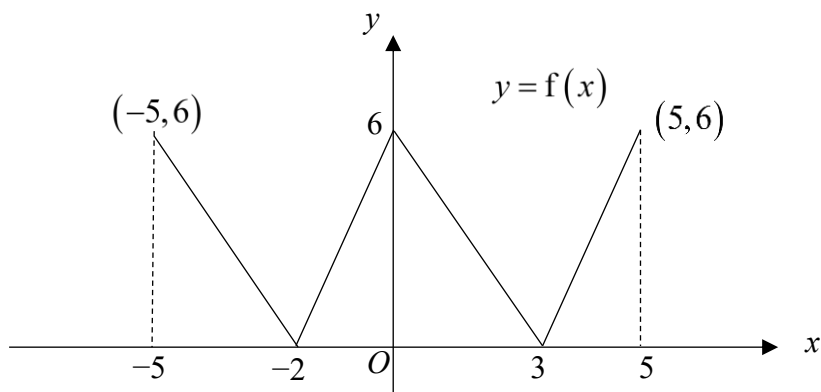
Common lines of symmetry for both  $C_1$  and  $C_2$  are  $x = 0$  and  $y = 0$ .

(b)

(i)

$$f(46) = f(41) = f(36) = \dots = f(1) = -2(1) + 6 = 4$$

(ii)



(iii)

For  $-5 \leq x \leq 5$ , the roots of  $f(-x) = 0$  are  $-3$  and  $2$ .

**Q6****(i)**

$$\begin{aligned}
& \int x^2 e^{-2x} dx \\
&= -\frac{1}{2} x^2 e^{-2x} + \int x e^{-2x} dx \\
&= -\frac{1}{2} x^2 e^{-2x} - \frac{1}{2} x e^{-2x} + \frac{1}{2} \int e^{-2x} dx \\
&= -\frac{1}{2} x^2 e^{-2x} - \frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} + C \\
&= -\frac{1}{4} e^{-2x} [2x^2 + 2x + 1] + C
\end{aligned}$$

$$\begin{aligned}
u &= x^2 & \frac{dv}{dx} &= e^{-2x} \\
\frac{du}{dx} &= 2x & v &= -\frac{1}{2} e^{-2x} \\
u &= x & \frac{dv}{dx} &= e^{-2x} \\
\frac{du}{dx} &= 1 & v &= -\frac{1}{2} e^{-2x}
\end{aligned}$$

**(ii)**

Volume

$$\begin{aligned}
&= \pi \int_0^1 (x e^{-x})^2 - \left( \frac{1}{e} \right)^2 dx \quad \text{or} \quad \pi \int_0^1 (x e^{-x})^2 dx - \frac{1}{3} \pi \left( \frac{1}{e} \right)^2 (1) \\
&= \pi \int_0^1 x^2 e^{-2x} - \frac{1}{e^2} x^2 dx \\
&= \pi \left[ -\frac{1}{4} e^{-2x} (2x^2 + 2x + 1) \right]_0^1 - \frac{\pi}{3e^2} [x^3]_0^1 \\
&= \pi \left[ -\frac{1}{4} e^{-2} (2(1)^2 + 2(1) + 1) \right] + \frac{\pi}{4} - \frac{\pi}{3e^2} (1)^3 \\
&= -\frac{5\pi}{4e^2} + \frac{\pi}{4} - \frac{\pi}{3e^2} \\
&= \frac{\pi}{4} - \frac{19\pi}{12e^2} \\
&= \frac{\pi}{12} (3 - 19e^{-2})
\end{aligned}$$

**Q7****(a)(i)**

$$\begin{aligned}
& \int \frac{x}{\sqrt{25-x^2}} dx \\
&= -\frac{1}{2} \int -2x (25-x^2)^{-\frac{1}{2}} dx \\
&= -\sqrt{25-x^2} + C
\end{aligned}$$

(a)(ii)

$$\begin{aligned}
& \int_{\alpha}^4 \left| \frac{x}{\sqrt{25-x^2}} \right| dx = 3 \\
& -\int_{\alpha}^0 \frac{x}{\sqrt{25-x^2}} dx + \int_0^4 \frac{x}{\sqrt{25-x^2}} dx = 3 \\
& -\left[-\sqrt{25-x^2}\right]_{\alpha}^0 + \left[-\sqrt{25-x^2}\right]_0^4 = 3 \\
& \left[\sqrt{25-x^2}\right]_{\alpha}^0 - \left[\sqrt{25-x^2}\right]_0^4 = 3 \\
& 5 - \sqrt{25-\alpha^2} - [3-5] = 3 \\
& \sqrt{25-\alpha^2} = 4 \\
& \alpha^2 = 9 \\
& \alpha = \pm 3 \\
& \text{Since } \alpha < 0, \alpha = -3
\end{aligned}$$

(b)

$$\begin{aligned}
x &= 4 \tan \theta \\
\frac{dx}{d\theta} &= 4 \sec^2 \theta \\
\text{When } x &= 4, \quad \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4} \\
\text{When } x &= 0, \quad \tan \theta = 0 \Rightarrow \theta = 0 \\
& \int_0^4 \sqrt{\frac{x^2}{16+x^2}} dx \\
&= \int_0^{\frac{\pi}{4}} \sqrt{\frac{16 \tan^2 \theta}{16+16 \tan^2 \theta}} 4 \sec^2 \theta d\theta \\
&= \int_0^{\frac{\pi}{4}} \sqrt{\frac{\tan^2 \theta}{\sec^2 \theta}} 4 \sec^2 \theta d\theta \\
&= 4 \int_0^{\frac{\pi}{4}} \frac{\tan \theta}{\sec \theta} \sec^2 \theta d\theta \quad \text{or} \quad = 4 \int_0^{\frac{\pi}{4}} \sin \theta \sec^2 \theta d\theta \\
&= 4 \int_0^{\frac{\pi}{4}} \tan \theta \sec \theta d\theta \quad = -4 \int_0^{\frac{\pi}{4}} -\sin \theta [\cos \theta]^{-2} d\theta \\
&= 4 \left[ \sec \theta \right]_0^{\frac{\pi}{4}} \quad = -4 \left[ \frac{(\cos \theta)^{-1}}{-1} \right]_0^{\frac{\pi}{4}} \\
&= 4 \left[ \sec \frac{\pi}{4} - \sec 0 \right] \quad = 4 \left[ \left( \cos \frac{\pi}{4} \right)^{-1} - (\cos 0)^{-1} \right] \\
&= 4(\sqrt{2}-1) \quad = 4(\sqrt{2}-1)
\end{aligned}$$

**Q8****(a)**

$$u_n = an^2 + bn + c$$

$$u_1 = a + b + c = 9 \text{-----} (1)$$

$$u_2 = 4a + 2b + c = 27 \text{----} (2)$$

$$u_3 = 9a + 3b + c = 55 \text{----} (3)$$

Using GC,  $a = 5, b = 3, c = 1$

$$u_n = 5n^2 + 3n + 1$$

**(b)(i)**

$$\begin{aligned} \sum_{r=1}^n (2r^3 + 5) &= 2 \sum_{r=1}^n r^3 + \sum_{r=1}^n 5 \\ &= 2 \left( \frac{n^2(n+1)^2}{4} \right) + 5n \\ &= \frac{n^2(n+1)^2}{2} + 5n \end{aligned}$$

**(b)(ii)****Method 1**

$$\begin{aligned} \sum_{r=2}^n (2(r+2)^3 + 5) &= (2(2+2)^3 + 5) + \dots + (2(n+2)^3 + 5) \\ &= \sum_{r=4}^{n+2} (2r^3 + 5) \\ &= \sum_{r=1}^{n+2} (2r^3 + 5) - \sum_{r=1}^3 (2r^3 + 5) \\ &= \frac{(n+2)^2(n+3)^2}{2} + 5(n+2) - \left[ \left( \frac{3^2(4)^2}{2} \right) + 5(3) \right] \\ &= \frac{(n+2)^2(n+3)^2}{2} + 5n - 77 \end{aligned}$$



**Method 2**

$$\begin{aligned}
\sum_{r=2}^n (2(r+2)^3 + 5) &= (2(2+2)^3 + 5) + \dots + (2(n+2)^3 + 5) \\
&= \sum_{r=4}^{n+2} (2r^3 + 5) \\
&= 2 \left[ \sum_{r=1}^{n+2} r^3 - \sum_{r=1}^3 r^3 \right] + \sum_{r=4}^{n+2} 5 \\
&= 2 \left[ \frac{(n+2)^2 (n+3)^2}{4} - \frac{3^2 (4)^2}{4} \right] + 5(n-1) \\
&= \frac{(n+2)^2 (n+3)^2}{2} + 5n - 77
\end{aligned}$$

**(b)(iii)**

From (i)  $\sum_{r=1}^n u_r = \frac{n^2 (n+1)^2}{2} + 5n$

As  $n \rightarrow \infty$ ,  $\frac{n^2 (n+1)^2}{2} \rightarrow \infty$ ,  $5n \rightarrow \infty$

Hence,  $\sum_{r=1}^n u_r = \frac{n^2 (n+1)^2}{2} + 5n \rightarrow \infty$ , series  $\sum_{r=1}^{\infty} u_r$  does not converge.

**Q9****(i)**

$$\int \frac{1}{4+9x^2} dx = \int \frac{1}{9\left(\frac{4}{9}+x^2\right)} dx = \frac{1}{6} \tan^{-1} \frac{3x}{2} + K$$

**(ii)**

$$\begin{aligned} f(x) &= (4+9x^2)^{-1} = \frac{1}{4} \left(1 + \frac{9x^2}{4}\right)^{-1} \\ &= \frac{1}{4} \left[ 1 + (-1) \left(\frac{9x^2}{4}\right) + \frac{-1(-2)}{2} \left(\frac{9x^2}{4}\right)^2 + \dots \right] \\ &= \frac{1}{4} \left( 1 - \frac{9x^2}{4} + \frac{81}{16} x^4 + \dots \right) \\ &\approx \frac{1}{4} - \frac{9}{16} x^2 + \frac{81}{64} x^4 \end{aligned}$$

**(iii)**

From **(i)**  $\tan^{-1} \frac{3x}{2} = 6 \int \frac{1}{4+9x^2} dx + C$ , where  $C = -6K$

From **(ii)**  $\tan^{-1} \frac{3x}{2} = 6 \int \left( \frac{1}{4} - \frac{9}{16} x^2 + \frac{81}{64} x^4 \right) dx + C$

$$\begin{aligned} &= 6 \left( \frac{1}{4} x - \frac{3}{16} x^3 + \frac{81}{320} x^5 \right) + D \\ &= \frac{3}{2} x - \frac{9}{8} x^3 + \frac{243}{160} x^5 + D \end{aligned}$$

$x = 0, \tan^{-1} 0 = 0 \Rightarrow D = 0$

$$\therefore \tan^{-1} \frac{3x}{2} = \frac{3}{2} x - \frac{9}{8} x^3 + \frac{243}{160} x^5$$

**(iv)**

$$\int_0^{0.5} \tan^{-1} \frac{3x}{2} dx = \int_0^{0.5} \left( \frac{3}{2} x - \frac{9}{8} x^3 + \frac{243}{160} x^5 \right) dx = 0.174 \text{ (to 3 dp)} \quad (\text{Using GC})$$

**(v)**

From GC,  $\int_0^{0.5} \tan^{-1} \frac{3x}{2} dx = 0.173 \text{ (to 3 dp)}$

**(vi)**

The estimate in **(iv)** is accurate up to 2 decimal places but not to 3 decimal places.

To improve the estimate, we can include higher-order terms in the Maclaurin series expansion of  $\tan^{-1} \frac{3x}{2}$ .

**Q10****(a)**

$$y = \ln\left(\frac{e^2}{3x}\right) = \ln e^2 - \ln 3x = 2 - \ln 3x$$

A

B

C

$$y = \ln x \rightarrow y = \ln 3x \rightarrow y = -\ln 3x \rightarrow y = 2 - \ln 3x$$

**A:** A scaling parallel to the  $x$ -axis by a factor of  $\frac{1}{3}$ .

**B:** Reflection in the  $x$ -axis.

**C:** A translation of 2 units in the positive direction of the  $y$ -axis.

**(b)(i)**

$$g(x) = 2f(x-1)$$

$$(a, b) \rightarrow (a+1, b) \rightarrow (a+1, 2b)$$

Hence the corresponding point  $R$  is  $(a+1, 2b)$ .

$$g'(x) = 2f'(x-1)$$

$$\text{Given } f'(a) = 5$$

$$f'(a+1) = 5 \quad [\text{gradient remains the same after translation}]$$

$$\text{At } (a+1, 2b),$$

$$g'(x) = 2f'(a+1) = 2(5) = 10$$

Alternative

When  $x = a+1$

$$g'(x) = 2f'(a+1-1) = 2f'(a)$$

$$\text{Given } f'(a) = 5,$$

$$g'(x) = 2(5) = 10$$

**(b)(ii)**

$$g(x) = \frac{1}{f(x)}$$

The corresponding point  $R$  is  $\left(a, \frac{1}{b}\right)$ .

$$g'(x) = -\frac{f'(x)}{[f(x)]^2}$$

$$\text{Given } f'(a) = 5, \text{ at } \left(a, \frac{1}{b}\right),$$

$$g'(x) = -\frac{f'(a)}{[f(a)]^2} = -\frac{5}{b^2}$$

Q11

(i)

$$u_{n+1} = (1+k)u_n$$

$$\frac{u_{n+1}}{u_n} = 1+k = \text{constant independent of } n \text{ (since } k \text{ is a constant),}$$

$\therefore \{u_n\}$  is a geometric progression.

(ii)

**Method 1**

$$u_1 = 311 \quad r = 1+k$$

$$S_3 = \frac{311((1+k)^3 - 1)}{1+k-1} = 4043 \quad \text{or} \quad \text{Using GC Table}$$

$$311(1+k)^3 - 311 = 4043k$$

$$(1+k)^3 - 13k - 1 = 0$$

$$k^3 + 3k^2 - 10k = 0$$

Using GC,  $k = -5, 0, 2$

Since  $k > 0$ ,  $k = 2$

**Method 2**

$$u_1 = 311 \quad u_2 = (1+k)311 \quad u_3 = (1+k)^2 311$$

$$311 + (1+k)311 + (1+k)^2 311 = 4043$$

$$1 + 1 + k + (1+k)^2 = 13$$

$$k^2 + 3k - 10 = 0$$

$$k = 2 \quad \text{or} \quad -5 \text{ (NA since } k > 0)$$

| X  | Y1     | Y2    |
|----|--------|-------|
| 0  | 0      | 0     |
| 1  | 2177   | 4043  |
| 2  | 8086   | 8086  |
| 3  | 19593  | 12129 |
| 4  | 38584  | 16172 |
| 5  | 66865  | 20215 |
| 6  | 106362 | 24258 |
| 7  | 158921 | 28301 |
| 8  | 226408 | 32344 |
| 9  | 310689 | 36387 |
| 10 | 413630 | 40430 |

(iii)

Since common ratio  $r = 3 > 1$ , the series does not converge as  $n \rightarrow \infty$ . Sum to infinity does not exist and hence there is no limit to total number of daily views in the long run.

(iv)

| $r$      | $v_r$                  |
|----------|------------------------|
| 1        | $v_1 = u_1 = 311$      |
| 2        | $v_2 = u_2 = (3)311$   |
| 3        | $v_3 = u_3 = (3^2)311$ |
| 4        | $v_4 = u_4 = (3^3)311$ |
| 5        | $v_5 = v_4 + 80$       |
| 6        | $v_6 = v_5 + 80$       |
| $\vdots$ |                        |

AP with 1<sup>st</sup> term  
 $(3^3)311 = 8397$ , common  
 difference 80

From  $v_4$  to  $v_r$ , no. of terms  $= r - 4 + 1 = r - 3$

$$\begin{aligned}
\therefore S_r &= S_3 + \frac{r-3}{2} [2(8397) + (r-4)(80)] \\
&= 4043 + 8397r - 25191 + 40(r-3)(r-4) \\
&= 40r^2 + 8117r - 20668 \quad \text{where } t = 20668 \quad (\text{shown})
\end{aligned}$$

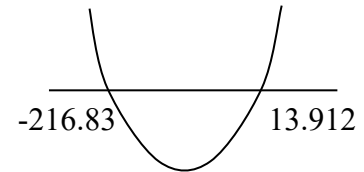
(v)

$$40r^2 + 8117r - 20668 > 100000$$

$$40r^2 + 8117r - 120668 > 0$$

$$r < -216.83 \text{ (NA since } r \geq 1) \quad \text{or} \quad r > 13.912$$

Least number of days = 14



Or using GC Table

| $r$ | $40r^2 + 8117r - 120668$ |
|-----|--------------------------|
| 13  | -8387 < 0                |
| 14  | 810 > 0                  |
| 15  | 10087 > 0                |

Least number of days = 14

**Q12****(i)**

Plane contains  $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$  and is parallel to  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$  and  $\begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$ .

Normal of plane,  $\mathbf{n} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}$

Equation of plane,

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}$$

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = -12$$

$$x - 4z = -12 \quad (\text{Shown})$$

**(ii)**

Normal of wall,  $\mathbf{n}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$  and normal of roof,  $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}$

Let  $\theta$  be acute angle between wall and roof.

$$\cos \theta = \frac{\left| \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} \right|}{(1)(\sqrt{17})}$$

$$= \frac{1}{\sqrt{17}}$$

$$\theta = 75.9638^\circ$$

Hence, obtuse angle between wall and roof  $= 180^\circ - 75.9638^\circ$

$$= 104.0362$$

$$\approx 104^\circ \quad (\text{nearest degree})$$

(iii)

**Method 1**

$$\overrightarrow{AP} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ -3 \end{pmatrix} \quad \text{or} \quad \overrightarrow{BP} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \\ -3 \end{pmatrix}$$

$$\text{Minimum length required} = \left| \overrightarrow{AP} \cdot \mathbf{n}_2 \right| \quad \text{or} \quad \left| \overrightarrow{BP} \cdot \mathbf{n}_2 \right|$$

$$= \frac{\left| \begin{pmatrix} 4 \\ 3 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} \right|}{\sqrt{17}} = \frac{16}{\sqrt{17}} \quad \text{or} \quad = \frac{\left| \begin{pmatrix} 4 \\ -2 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} \right|}{\sqrt{17}} = \frac{16}{\sqrt{17}}$$

**Method 2: (Finding foot of perpendicular from  $P$  to roof)**

$$\text{Eq of roof: } \mathbf{r} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = -12 \quad \text{and} \quad \text{Eq of wooden strut: } \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$\text{Then, } \begin{pmatrix} 4 + \lambda \\ 3 \\ -4\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = -12$$

$$4 + \lambda + 16\lambda = -12$$

$$17\lambda = -16$$

$$\lambda = -\frac{16}{17}$$

$$\text{Length of wooden strut} = \left| \overrightarrow{PF} \right| = \left| \overrightarrow{OF} - \overrightarrow{OP} \right|$$

$$= \left| \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} - \frac{16}{17} \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} \right|$$

$$= \left| \begin{pmatrix} -\frac{16}{17} \\ 0 \\ \frac{64}{17} \end{pmatrix} \right|$$

$$= \sqrt{\left(-\frac{16}{17}\right)^2 + \left(\frac{64}{17}\right)^2}$$

$$= \frac{16}{\sqrt{17}}$$

(iv)

Equation of wooden strut:  $\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix}, \quad \lambda \in \mathbb{R}$

Equation of light ray:  $\mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ -4 \\ 4 \end{pmatrix}, \quad \mu \in \mathbb{R}$

Assuming they intersect at  $X$ ,

$$\overrightarrow{OX} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 7 \\ -4 \\ 4 \end{pmatrix}, \quad \text{for some } \lambda, \mu \in \mathbb{R}$$

$$4 + \lambda = 0 + 7\mu \quad -(1) \Rightarrow \quad \lambda - 7\mu = -4 \quad -(4)$$

$$3 = 5 - 4\mu \quad -(2) \Rightarrow \quad \mu = \frac{1}{2}$$

$$-4\lambda = 4\mu \quad -(3) \Rightarrow \quad \lambda = -\mu = -\frac{1}{2}$$

Substitute  $\lambda = -\frac{1}{2}$  and  $\mu = \frac{1}{2}$  into (4):  $\text{LHS} = \lambda - 7\mu = -\frac{1}{2} - 7\left(\frac{1}{2}\right) = -4 = \text{RHS}$

Hence,  $\lambda = -\frac{1}{2}$  and  $\mu = \frac{1}{2}$  satisfy all 3 equations.

The light ray meets the strut.

$$\text{And } \overrightarrow{OX} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} + \left(-\frac{1}{2}\right) \begin{pmatrix} 1 \\ 0 \\ -4 \end{pmatrix} = \begin{pmatrix} 3.5 \\ 3 \\ 2 \end{pmatrix} \quad \text{or} \quad \overrightarrow{OX} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} + \left(\frac{1}{2}\right) \begin{pmatrix} 7 \\ -4 \\ 4 \end{pmatrix} = \begin{pmatrix} 3.5 \\ 3 \\ 2 \end{pmatrix}$$

The light ray meets the wooden strut at  $(3.5, 3, 2)$ .